

Sales Growth and Profit Quality

Superstore Business Analysis Report

Executive Summary

From 2023 to 2026, the business generated \$2.33M in sales and \$292.30K in profit at a 12.56% weighted margin. Management's concern is supported: sales and profit growth do not consistently move together. Sales declined 4.26% in 2024 while profit increased 20.00%; in 2026, sales grew 21.44% while profit grew only 16.04%. The main leakage is concentrated in Furniture, Central, and discounts of 30% or more. The recommended response is targeted margin governance rather than broad sales suppression.

1. Business Context and Questions

Management observed that changes in sales scale did not consistently translate into profit growth. This project assesses the quality of growth and locates where margin is being eroded. It asks whether the portfolio is healthy overall, when sales and profit diverge, which categories and regions explain the divergence, and whether discounts, returns, or shipping provide useful diagnostic context.

2. Data and Methodology

Item	Definition	Result
Source	Sample Superstore: Orders, Returns, People	10,194 order lines
Period	Order Date	2023-01-03 to 2026-12-30
Orders	COUNTD(Order ID)	5,111
Customers	COUNTD(Customer ID)	804
Profit Margin	SUM(Profit) / SUM(Sales)	12.56%
Product key	Product ID + Product Name	Required due to mapping issues

Validation found no missing critical fields, duplicate Row IDs, shipments preceding orders, invalid sales/quantity/discount values, or orphan records in the Returns and People joins. Product ID and Product Name are not strictly one-to-one, so product analysis uses a composite key.

3. Overall Performance

KPI	Value	Interpretation
Sales	\$2,326,534.35	Large portfolio scale across four years
Profit	\$292,296.81	Positive overall
Profit Margin	12.56%	Healthy baseline, but uneven by business mix

KPI	Value	Interpretation
Orders	5,111	Distinct orders
Customers	804	Broad customer base

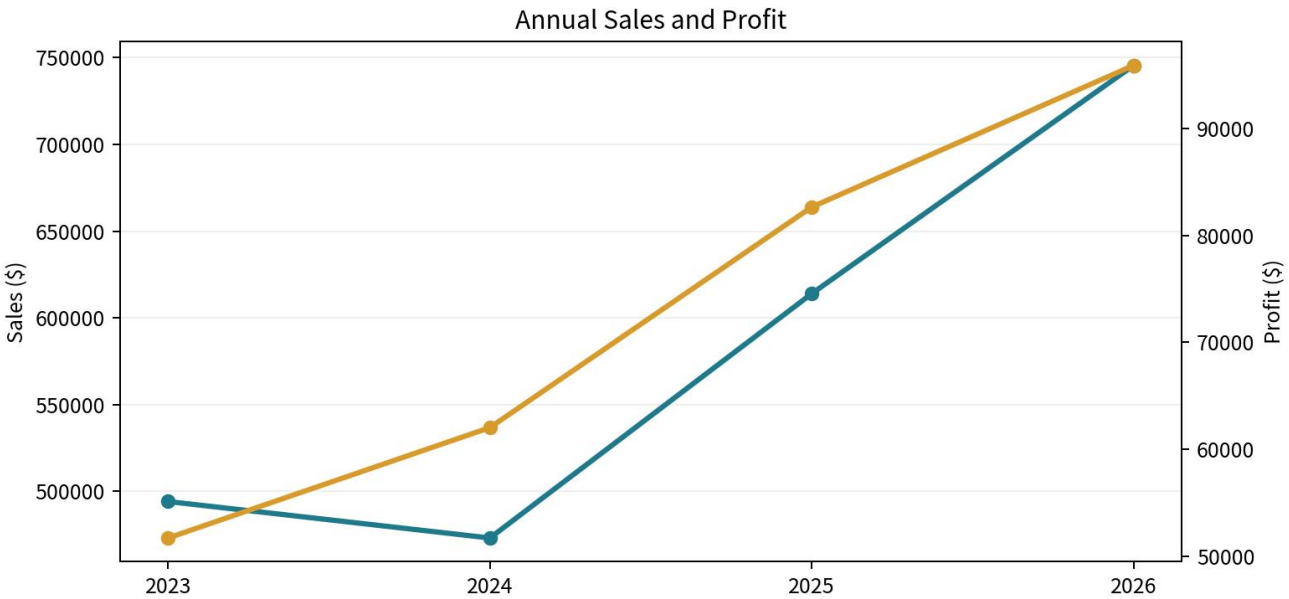


Figure 1. Annual sales and profit follow the same broad direction, but growth rates diverge.

4. Key Findings

4.1 Growth is not uniformly profit-accretive

In 2024, sales declined 4.26% while profit increased 20.00%, lifting margin from 10.46% to 13.11%. In 2026, sales grew 21.44% but profit grew only 16.04%, and margin declined to 12.87%. Product, regional, and discount mix therefore materially affect how efficiently sales convert into profit.

4.2 Furniture is the structural margin drag

Furniture generated \$754.7K in sales, or 32.44% of the total, but only \$19.7K in profit, or 6.75% of total profit. Its margin was 2.61%, compared with 17.45% for Technology and 17.22% for Office Supplies. The issue is not weak sales volume; it is poor conversion of a large revenue base into profit.

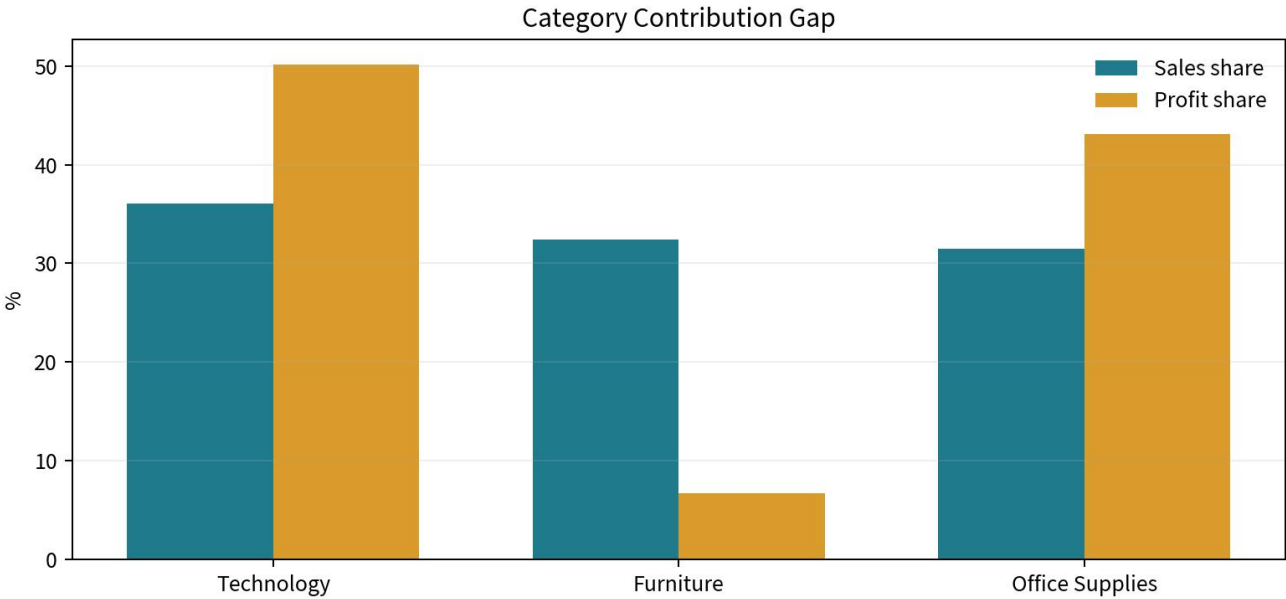


Figure 2. Furniture shows the largest gap between sales contribution and profit contribution.

4.3 Central is the priority region

Central generated \$503.2K in sales, representing 21.63% of the portfolio, but achieved only a 7.92% margin. Central Furniture lost \$2,802. Within that category, Furnishings, Tables, and Bookcases lost approximately \$3,916, \$3,560, and \$1,998, respectively. Appliances, Binders, and Supplies were also loss-making within Office Supplies.

4.4 Discount is the clearest actionable lever

Lines with no discount achieved a 29.56% aggregate margin, and a 20% discount still produced an 11.77% margin. At a 30% discount, margin turned negative at -10.06% and generally deteriorated at deeper levels. This is observational rather than causal evidence, but it is sufficient to establish a review threshold at 30% and prioritize combinations such as Central × Furniture.

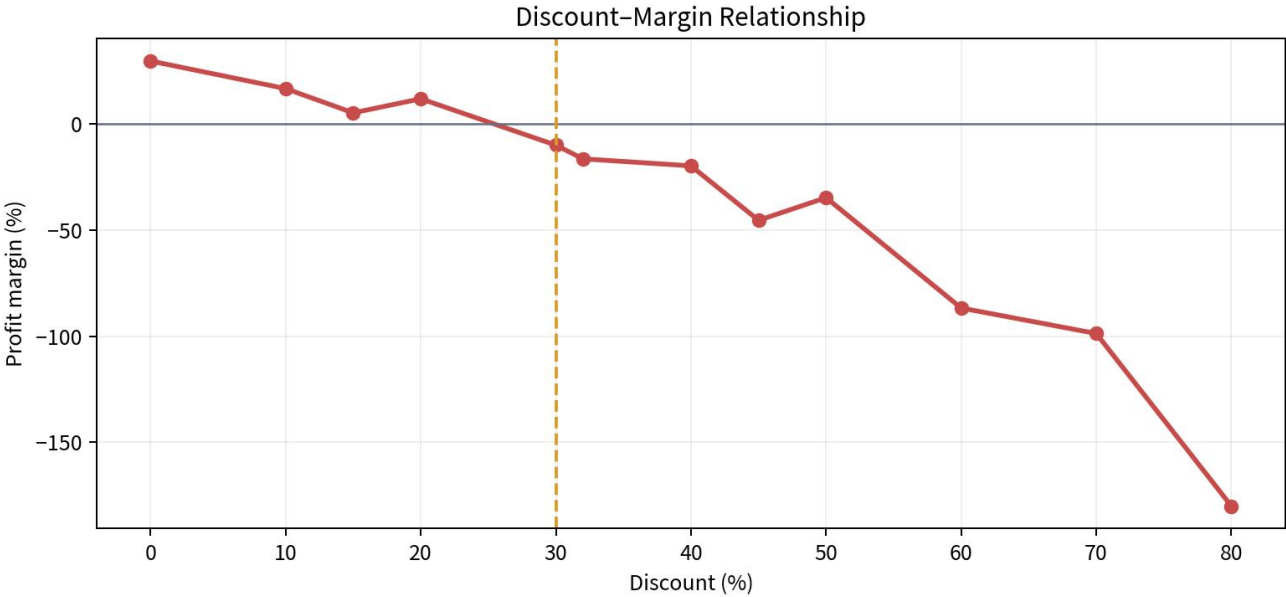


Figure 3. Aggregate margin turns negative at a 30% discount and deteriorates thereafter.

4.5 Returns are a separate West issue

West's return rate was 11.56%, far above Central at 3.31%, East at 2.98%, and South at 2.92%. Returns therefore do not explain Central's weak margin, but West warrants a separate order- and product-level return investigation. Shipping-mode margin differences were small and are treated as descriptive context, not causal evidence.

5. Recommendations

Priority	Action	Metric	Rationale
P1	Introduce approval or warning rules for discounts of 30% or more, calibrated by category	Margin and loss sales at $\geq 30\%$	Observed break-even warning point
P1	Review Central $\times$ Furniture, starting with Furnishings, Tables, and Bookcases	Profit, margin, discount mix	Largest concentrated leakage
P2	Monitor high-sales loss-making products using Product ID + Product Name	Loss amount, orders, repeat customers	Avoids broad category cuts
P2	Investigate West returns by product, customer, month, and ship mode	Return rate and returned sales	Separate operational exception
P3	Present Sales, Profit, Margin, and Growth Gap together in management reporting	Growth gap, weighted margin	Avoids revenue-only decisions

6. Limitations and Next Steps

- The dataset excludes procurement cost, list price, pre-discount price, promotion metadata, inventory, and customer contract terms, so a strict causal discount model cannot be built.
- Profit is a provided outcome field and cannot be decomposed into logistics, procurement, or return-processing cost components.
- Next steps include discount control-group comparisons, returned-order value analysis, customer lifecycle and repeat-purchase analysis, and integration with inventory and cost data in a real business environment.

Conclusion

The core conclusion is not that sales growth is undesirable, but that growth must be evaluated together with margin and business mix. The portfolio is profitable overall, yet Furniture, Central, and discounts of 30% or more form identifiable profit leakage. The appropriate response is targeted discount governance and loss-combination monitoring while preserving healthy growth in Technology and Office Supplies.